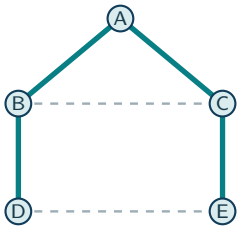
GRAPH THEORY LADDER 2 OF 9

Paths, cycles, connectivity and trees

Walks in a graph lead to cycles, and a connected graph with no cycles is a tree with one edge fewer than vertices.

1 / THE RULE AND THE PICTURE

A tree on n vertices has exactly $n-1$ edges



a spanning tree of the house graph in teal

A tree is a connected graph with no cycle, and removing the two dashed edges from the house leaves a tree through all 5 vertices, a spanning tree:

$V - E = 1$ for a tree, $\quad E - V + 1 =$ number of independent cycles

The house graph has $E - V + 1 = 6 - 5 + 1 = 2$ independent cycles, the triangle and the square, and removing one edge from each leaves a tree with 4 edges.

2 / THE WORDS FOR MOVING ABOUT

Each word allows fewer repeats than the one before

word	meaning	in the house graph
walk	vertices and edges in sequence, repeats allowed	A, B, A, C
trail	a walk with no repeated edge	B, A, C, B, D
path	no repeated vertex	A, B, D, E of length 3
cycle	a closed path of at least 3 vertices	A, B, C (a triangle) or B, D, E, C (a square)
connected	a path joins every pair of vertices	the house graph is connected

3 / WHY A TREE HAS $n-1$ EDGES

Peel off a leaf and count

Every tree with at least two vertices has a leaf, a vertex of degree 1, since a longest path must end at one; removing the leaf and its edge leaves a smaller tree, so by repetition the edges number one fewer than the vertices.
Two more facts follow: any two vertices of a tree are joined by exactly one path, and adding any new edge to a tree creates exactly one cycle.

4 / HOW MANY SPANNING TREES

The house graph has 11 and K_n has n^{n-2}

A spanning tree chooses which edges to keep, and for the house graph there are 11 ways, which the Laplacian sheet later counts with a determinant.
Cayley's formula says the complete graph K_n has n^{n-2} spanning trees, so K_4 has 16 and K_5 has 125, the same numbers as the labelled trees on four and five vertices, since every labelled tree is a spanning tree of K_n.

5 / TRY THESE

Find the cycles and the tree

(a) List the cycles of the house graph: the triangle ABC, the square $BDEC$ and the pentagon $ABDEC$.
(b) How many edges must be removed from K_4 to leave a tree? $6 - 3 = 3$ of them.
(c) A graph with 10 vertices and 9 edges that is connected must be a tree, because it has $V - 1$ edges.

ASIDE / FORESTS

A graph with no cycles that is not connected is a forest, and a forest with c components on n vertices has $n - c$ edges.

LINKS TO OTHER SHEETS

The first sheet gives the degree rule used here; the Euler formula sheet turns the cycle count $E - V + 1$ into a count of faces.

Convention. A spanning tree uses all vertices; $E - V + 1$ is the cycle rank of a connected graph; Cayley counted labelled trees.